

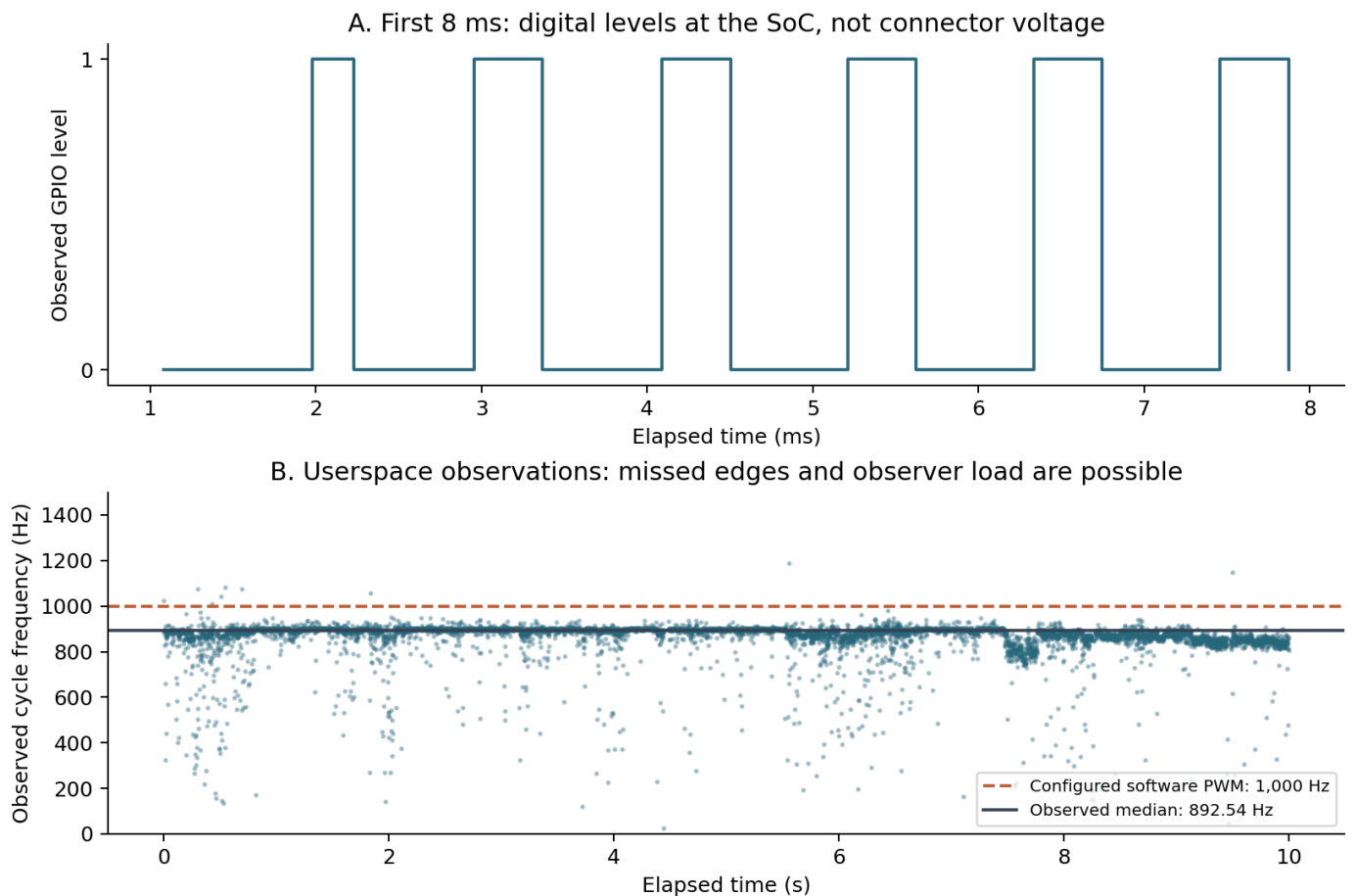
pifanctl: Hardware Verification

NF-A12x25 PWM / Pi 4 shared-fan rack / 1.1.0 runtime

INTERIM REPORT: PHYSICAL ACCEPTANCE NOT ACHIEVED

This report records a bounded read-only GPIO observation. It does not certify connector-level PWM, electrical safety, measured RPM, airflow or cooling capacity.

Observation	Result
Duration / complete cycles	10.00001 s / 8,432
Configured / median observed frequency	1,000 Hz / 892.54 Hz
Commanded / median observed HIGH ratio	35.04% / 36.59%
Median / p99 sampling gap	2.95 / 7.56 microseconds
Maximum sampling gap	38.28 ms: missed edges possible
Tach pulses / existing bias	0 / pull-down: inconclusive
Pin configuration / worker	Unchanged / Ready, no restart



Clock: userspace monotonic transition observations. Long gaps and CPU observer effects prevent calibrated jitter claims. SoC levels do not establish cable continuity or connector voltage.

Remaining procedure and acceptance gates

01. Electrical inspection

Confirm supply, common ground, tach continuity and Pi-safe voltage. Identify adapters and any external pull-up. Do not enable bias until the input is verified.

02. Conforming PWM

Noctua specifies a 25 kHz target, acceptable 21-28 kHz. Validate the actual connector voltage, duty, polarity and frequency. The current software 1 kHz path does not establish this contract; no kernel PWM chip is currently exposed to the worker.

03. Hardware PWM handoff

Review the Pi 4 overlay and actual chip/channel mapping. Preserve single-writer ownership, independent cooling, resource identity and rollback. A hardware driver change requires controlled immutable-Fan replacement. Do not guess channels or relabel software PWM as hardware PWM.

04. Tach baseline

After wiring passes, enable optional feedback with the verified BCM input, appropriate 3.3 V bias and two pulses/revolution. Record 60 s of fresh pulse counts, RPM, requested duty and member temperatures without changing cooling policy.

05. Duty/RPM sweep

Use one managed writer for 100%, 75%, 50% and 30% stages. Settle at least 30 s, accounting for the falling-duty limiter, and observe 60 s per stage. Verify an independent local deadline/temperature guard first. Abort at any member reaching 60 C, stale telemetry, collector errors or lost observation. Restore normal policy on every exit.

06. Broader physical scope

Startup/shutdown, reboot, power loss, network partition, Pi 5 actuation, full v0 rollback and larger-fleet capacity remain unverified. They require their own physical setup and safeguards. Existing failed thermal holds remain unchanged.

Acceptance ledger: GPIO observation completed. Connector PWM, usable tach baseline, duty/RPM response and the broader physical matrix remain NOT MEASURED / NOT RUN. No fan stop, load trial or reduced-duty sweep was performed here.

Sources and reproducibility

Noctua PWM specifications:

https://www.noctua.at/pub/media/wysiwyg/Noctua_PWM_specifications_white_paper.pdf

Linux PWM interface: <https://docs.kernel.org/driver-api/pwm.html>

Project tracking: github.com/jyje/pifanctl/issues/69 and [/64](https://github.com/jyje/pifanctl/issues/64)

Anonymous CSV/summary and the complete procedure are in docs/v1/hardware-verification/. Rebuild with scripts/build_hardware_observation.py. The collector is read-only, bounded and specific to the reviewed Pi 4 pin mapping.